

TechBiz Security

Cyber Security Internship Program

CYBER SECURITY INTERNSHIP REPORT

Week 4 – Task 10 (EH)

Vulnerability Scanning Basics

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1. Introduction

Vulnerability scanning is the process of identifying potential security weaknesses in systems, services, applications, and network configurations using automated tools and vulnerability detection techniques. It forms one of the earliest and most important stages of a security assessment, allowing an assessor to build a picture of the attack surface before deeper analysis or testing is carried out.

Within a broader penetration testing or vulnerability assessment workflow, vulnerability scanning typically follows reconnaissance and service enumeration, and precedes validation, exploitation (where authorised), and reporting. Automated scanning tools compare the characteristics of discovered services against known vulnerability signatures, giving an assessor an initial, prioritised list of areas that require closer review.

For this task, Nmap Scripting Engine (NSE) vulnerability detection scripts were used against an authorised Metasploitable 2 laboratory target. It should be noted that Metasploitable 2 is an intentionally vulnerable virtual machine built for training purposes, and the findings documented in this report reflect a controlled lab environment rather than a production system.

2. Objective

- Understand the fundamentals of automated vulnerability scanning.
- Perform vulnerability detection using Nmap NSE scripts.
- Identify potential vulnerabilities from scan results.
- Map detected vulnerabilities to their corresponding CVE identifiers.
- Research vulnerability descriptions and severity information.
- Understand how vulnerability databases such as NVD can be used during security assessments.
- Complete the mandatory TryHackMe Vulnerabilities 101 learning activity.
- Document the findings professionally.
- Recommend appropriate remediation measures.

3. Lab Environment

Target	Metasploitable 2
Target IP	10.209.222.43
Attacker / Assessment Machine	Kali Linux
Scanning Tool	Nmap
Nmap Function	NSE vulnerability detection scripts
Command Used	<code>nmap --script vuln 10.209.222.43</code>
Assessment Type	Authorized laboratory vulnerability assessment

Metasploitable 2 is an intentionally vulnerable Linux virtual machine that is widely used for cybersecurity education and penetration-testing practice. It is deliberately configured with outdated and misconfigured services so that learners can safely practice vulnerability identification and assessment techniques in an isolated, authorised environment.

4. Tools and Technologies Used

Kali Linux: Used as the cybersecurity assessment environment from which all scanning activity was performed.

Nmap: Network scanning and service enumeration tool used to identify open services and perform vulnerability detection through NSE scripts.

Nmap NSE: The Nmap Scripting Engine, used to automate security checks and identify known vulnerabilities associated with discovered services.

NVD: The National Vulnerability Database, used to research CVE information, vulnerability descriptions, and severity data.

Rapid7: Used as a supplementary reference for CVE-2011-2523 and the vsFTPD 2.3.4 backdoor information.

TryHackMe: Used to complete the mandatory Vulnerabilities 101 learning activity and understand vulnerability concepts, CVSS, vulnerability databases, and vulnerability research.

5. Vulnerability Scanning Methodology

Step 1: Identify the authorized Metasploitable 2 target IP address.

Step 2: Run the Nmap vulnerability detection command:

```
nmap --script vuln 10.209.222.43
```

Step 3: Review the Nmap output for services marked as vulnerable or potentially vulnerable.

Step 4: Identify relevant vulnerability findings containing CVE identifiers.

Step 5: Research the identified CVEs using trusted vulnerability databases such as NVD and other authoritative security references.

Step 6: Record the vulnerability name, affected service, CVE identifier, description, severity information, security impact, and recommended remediation.

Automated scanning provides an initial assessment of potential weaknesses. Findings generated by tools such as Nmap NSE should be validated and interpreted through further research before being treated as confirmed security issues in a real environment.

6. Nmap Vulnerability Scan

Nmap NSE vulnerability scripts were executed against the authorized target, 10.209.222.43, using the following command:

```
nmap --script vuln 10.209.222.43
```

The scan identified multiple potential vulnerabilities and security weaknesses associated with services running on the Metasploitable 2 system.

The screenshot above shows the target IP address (10.209.222.43) and the Nmap vulnerability scanning command that was executed. The scan results identify the FTP service running on TCP port 21 and report detection of the vsFTPD 2.3.4 backdoor, referenced as CVE-2011-2523. The output also shows detection of additional TLS-related vulnerabilities, including an anonymous Diffie-Hellman key exchange weakness and a DHE_EXPORT cipher downgrade issue associated with the Logjam vulnerability. This output represents the result reported by the Nmap NSE vulnerability detection scripts and does not represent manual exploitation performed during this task.

Finding	Affected Service	CVE	Nmap Detection	Severity	Security Impact
vsFTPD 2.3.4 Backdoor	FTP – TCP/21	CVE-2011-2523	VULNERABLE (Exploitable)	High risk (based on finding nature)	Unauthorized command execution; possible system compromise
SSL POODLE	SSL/TLS service	CVE-2014-3566	VULNERABLE	CVSS 3.1: 3.4 – LOW	Potential information disclosure via SSL 3.0 padding-oracle attack
TLS DHE_EXPORT / Logjam	TLS service	CVE-2015-4000	VULNERABLE	CISA-ADP CVSS 3.1: 3.7 – LOW	Cryptographic downgrade; potential confidentiality impact

Nmap identified vsFTPD version 2.3.4 as containing the known backdoor associated with CVE-2011-2523. The Nmap output showed the NSE vulnerability detection script executing the “id” command against the service and returning “uid=0(root) gid=0(root)”. This result is described strictly as the output reported by the Nmap vulnerability detection script; no manual exploitation or interactive shell access was performed or documented as part of this task.

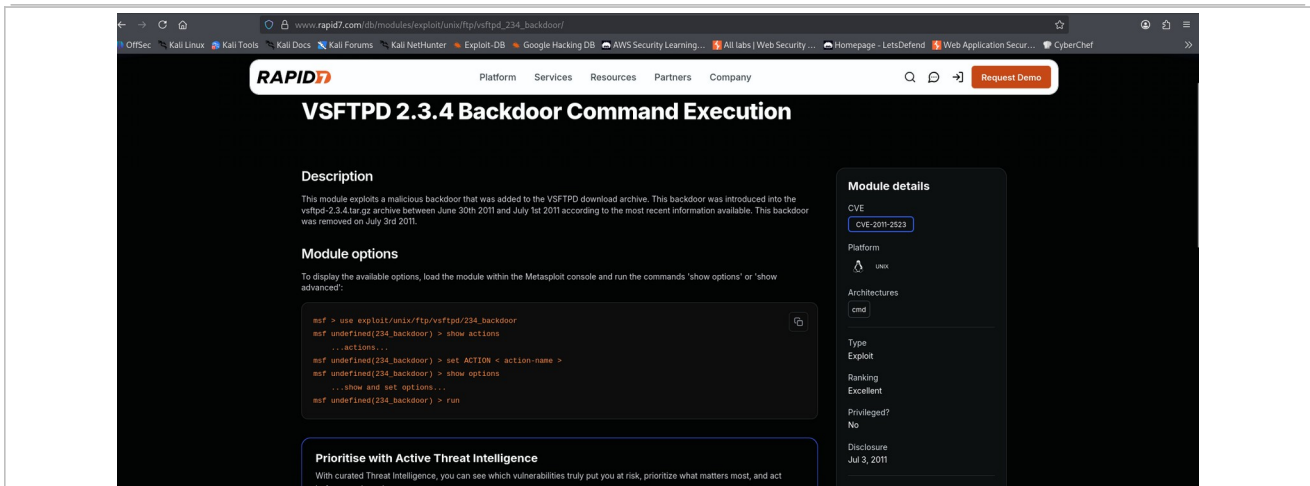


Figure 2: Reference information for the vsFTPd 2.3.4 backdoor vulnerability, CVE-2011-2523.

Security Impact: A compromised or backdoored FTP service can potentially allow unauthorized command execution and severe system compromise. Because the affected service is FTP and the vulnerability is associated with a malicious backdoor embedded in the software distribution, it represents a serious security concern on any system where this version remains deployed.

Recommended Remediation:

- Remove the vulnerable / backdoored vsFTPd version.
- Upgrade to a trusted, supported version obtained from a verified source.
- Disable unnecessary FTP services.
- Prefer secure file-transfer protocols such as SFTP where appropriate.
- Restrict network access to administrative services.
- Review the system for signs of compromise if such a vulnerable package was previously deployed.

7.2 Finding 2 – SSL POODLE

CVE-2014-3566 is associated with the POODLE vulnerability affecting SSL 3.0, resulting from weaknesses in CBC padding handling. This can allow an attacker positioned appropriately in a man-in-the-middle scenario to recover information from encrypted communications through a padding-oracle attack.

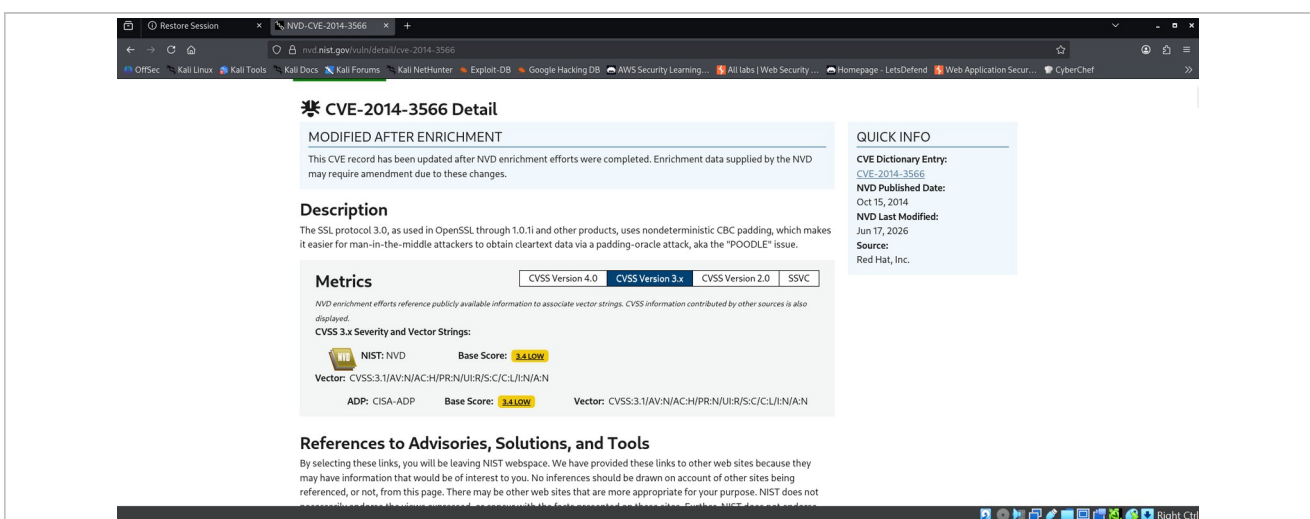


Figure 3: NVD vulnerability information for CVE-2014-3566 (SSL POODLE).

The NVD entry shown in the screenshot lists a CVSS 3.1 Base Score of 3.4, rated LOW. This finding should not be classified as Critical or High severity solely on the basis that Nmap flagged the service as vulnerable; the authoritative NVD severity rating for this CVE is LOW.

Recommended Remediation:

- Disable SSL 3.0.
- Use modern TLS versions.
- Disable obsolete and insecure cryptographic protocols and cipher configurations.
- Maintain secure TLS configurations in line with current security standards.

7.3 Finding 3 – TLS DHE_EXPORT / Logjam

The Logjam vulnerability, CVE-2015-4000, is associated with TLS configurations that permit DHE_EXPORT downgrade behaviour. This allows an attacker in a man-in-the-middle position to downgrade the cryptographic exchange to weaker export-grade Diffie-Hellman parameters.

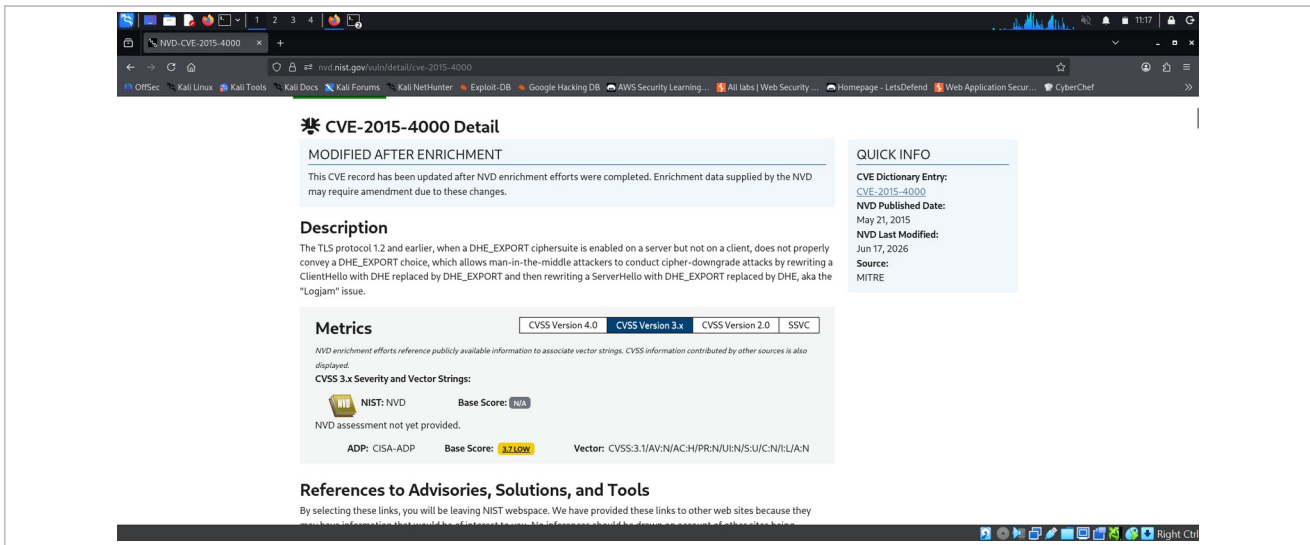


Figure 4: NVD vulnerability information for CVE-2015-4000 (Logjam).

The NVD screenshot shows the CVE was published on May 21, 2015 and last modified on June 17, 2026. The page indicates that an NVD assessment has not yet been provided, while the CISA-ADP CVSS 3.1 Base Score is listed as 3.7, rated LOW. This score should not be described as Medium severity, as the screenshot explicitly displays a LOW rating from CISA-ADP.

Recommended Remediation:

- Disable export-grade Diffie-Hellman cipher suites.
- Disable weak DHE_EXPORT configurations.
- Use strong, modern TLS configurations.
- Review and harden cryptographic settings.
- Keep TLS implementations and server software updated.

8. CVE Research and Analysis

A CVE (Common Vulnerabilities and Exposures) identifier is a standardized reference assigned to a publicly known cybersecurity vulnerability. CVE identifiers allow security professionals, vendors, and researchers to refer to the same vulnerability consistently across different tools, reports, and databases.

The National Vulnerability Database (NVD) is a U.S. government repository of vulnerability data that expands on CVE entries with additional information. Security professionals use vulnerability databases such as NVD to investigate vulnerability descriptions, affected products, CVSS scores, references, security impact, and remediation information during an assessment.

8.1 CVE-2011-2523

CVE-2011-2523 refers to a malicious backdoor that was inserted into the vsFTPD 2.3.4 source archive between June 30 and July 1, 2011, before being removed on July 3, 2011. Any deployment that installed the compromised archive during this window would contain a backdoor allowing command execution. The Rapid7 reference page shown previously in Figure 2 identifies this finding as “VSFTPD 2.3.4 Backdoor Command Execution”, associated with CVE-2011-2523.

8.2 CVE-2014-3566

CVE-2014-3566 (POODLE) affects SSL 3.0 due to non-deterministic CBC padding handling, which can allow an attacker to obtain cleartext data through a padding-oracle attack in a man-in-the-middle position. The NVD listing shown previously in Figure 3 records a CVSS 3.1 Base Score of 3.4, rated LOW.

8.3 CVE-2015-4000

CVE-2015-4000 (Logjam) affects the TLS protocol where a DHE_EXPORT ciphersuite is enabled on a server but not correctly conveyed to the client, allowing an attacker to conduct a cipher-downgrade attack. The NVD page shown previously in Figure 4 states that an NVD assessment has not yet been provided, while the CISA-ADP CVSS 3.1 Base Score is listed as 3.7, rated LOW.

9. TryHackMe Vulnerabilities 101

The mandatory TryHackMe “Vulnerabilities 101” room was completed as part of the learning component of this task. The room covered concepts including vulnerability fundamentals, vulnerability scoring, CVSS and VPR concepts, vulnerability databases, vulnerability research, and finding and understanding vulnerabilities.

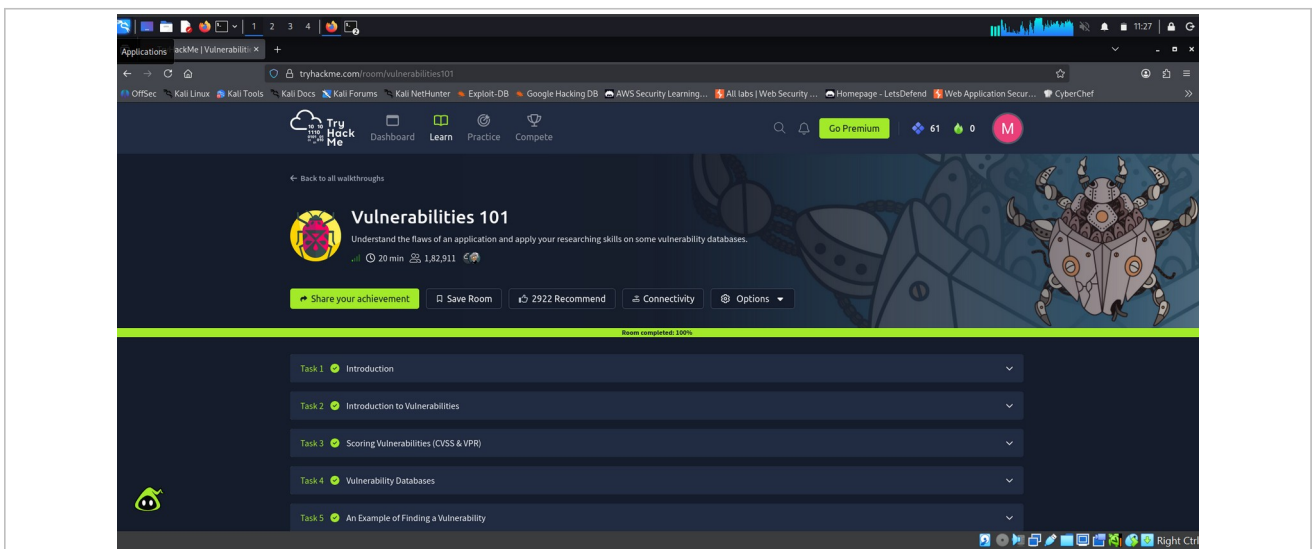


Figure 5: TryHackMe Vulnerabilities 101 room completed with 100% progress.

The screenshot above demonstrates 100% completion of the TryHackMe Vulnerabilities 101 room.

10. Risk and Severity Analysis

Vulnerability	CVE	Detection	Severity Information	Potential Impact
vsFTPd 2.3.4 Backdoor	CVE-2011-2523	Nmap VULNERABLE (Exploitable)	High-risk system compromise potential	Unauthorized command execution and possible system compromise
SSL POODLE	CVE-2014-3566	Nmap VULNERABLE	CVSS 3.1: 3.4 LOW	Potential information

Vulnerability	CVE	Detection	Severity Information	Potential Impact
				disclosure through exploitation of SSL 3.0 weakness
Logjam	CVE-2015-4000	Nmap VULNERABLE	CISA-ADP CVSS 3.1: 3.7 LOW	Cryptographic downgrade and potential confidentiality impact in vulnerable TLS configurations

No official severity rating was published for CVE-2011-2523 in the sources reviewed for this task. Its practical risk is therefore described based on the nature of the vulnerability itself and the Nmap “VULNERABLE (Exploitable)” detection result, rather than a formal CVSS score.

Severity scores are only one part of an overall risk assessment. Actual organizational risk also depends on factors such as asset importance, exposure to untrusted networks, ease of exploitability, the presence of compensating controls, and the potential business impact of a successful compromise.

11. Recommended Remediation

1. Remove or replace vulnerable vsFTPD 2.3.4 installations.
2. Use trusted and supported software packages.
3. Disable unnecessary FTP services.
4. Prefer secure file-transfer mechanisms such as SFTP.
5. Disable SSL 3.0.
6. Disable weak and export-grade cryptographic configurations.
7. Harden TLS configurations.
8. Apply security updates and patches.
9. Restrict access to exposed services using firewall and network segmentation controls.
10. Perform regular vulnerability scans.
11. Validate vulnerability findings before remediation closure.
12. Maintain an inventory of software versions and exposed services.

12. Practical Learning Outcomes

- Practical use of Nmap NSE vulnerability scripts.
- Understanding automated vulnerability detection.
- Identifying vulnerabilities from scan output.
- Understanding CVE identifiers.
- Researching vulnerabilities using NVD and security references.
- Understanding CVSS-based severity information.
- Understanding SSL/TLS security weaknesses.
- Understanding the importance of validating automated scan results.
- Documenting vulnerabilities in a professional security report.
- Understanding the difference between vulnerability scanning and exploitation.

13. Conclusion

This task provided practical exposure to vulnerability scanning using Nmap NSE against an authorized Metasploitable 2 laboratory system. The scan identified multiple security weaknesses, including the vsFTPD 2.3.4 backdoor (CVE-2011-2523), the SSL POODLE vulnerability (CVE-2014-3566), and the Logjam / DHE_EXPORT downgrade vulnerability (CVE-2015-4000).

Subsequent CVE research using the National Vulnerability Database and the Rapid7 reference, together with completion of the TryHackMe Vulnerabilities 101 room, strengthened understanding of vulnerability identification, vulnerability databases, severity assessment, and remediation planning. No exploitation of the identified vulnerabilities was performed as part of this task; the exercise was limited to automated vulnerability detection and research.

14. References

- Nmap NSE Documentation
- Nmap NSE Vulnerability Scripts Documentation
- National Vulnerability Database (NVD)
- Rapid7 Vulnerability / Exploit Reference for vsFTPD 2.3.4 Backdoor
- TryHackMe – Vulnerabilities 101